
FedCPR: Contrastive Learning with Prototype Rejection for Federated Open Set Recognition

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Abstract

Federated Learning (FL) typically operates under a closed-set assumption, where the label space is static and known across all clients. However, real-world deployments inevitably encounter "unknown" classes not seen during training, posing a significant challenge to model reliability. Existing methods struggle to distinguish these open-set samples, often classifying them over-confidently as known classes due to the lack of outlier exposure during local training. To address this, we propose a novel Federated Open-Set Recognition framework called FedCPR that makes use of Supervised Contrastive Learning with Curriculum-based Latent Prototype Rejection. Unlike traditional approaches that rely on raw pixel augmentation, our method perturbs intermediate feature representations to generate "hard" synthetic outliers. We enforce an entropy maximization objective on these augmented latents, effectively pushing them away from class prototypes and reserving low-density regions of the embedding space for actual unknown samples. Furthermore, we implement a curriculum schedule—progressing from structural to semantic perturbations—to stabilize training and prevent gradient conflicts during early convergence. Extensive experiments on partitioned CIFAR-10 demonstrate that our approach achieves SOTA results in open-set detection metrics (AUROC) and closed-set accuracy. The code for our proposed method can be found [here](#).

1. Introduction

Deep learning models typically require large amounts of data to achieve state-of-the-art performance. However, in practical deployment scenarios, ideal training conditions are rarely met. Data is often decentralized across multiple institutions and unavailable for aggregation due to strict privacy regulations and legal constraints. Furthermore, deployed models frequently encounter samples from distributions or

classes not present during training data they are not able to recognize. To address these concerns, Federated Learning (FL) emerged for training distributed models without sharing user data over the cloud, and furthermore, a need for recognising when a model was ill-equipped to conceive this data was formed.

Despite FL's popularity, the vast majority of work in federated settings operate under an ideal scenario via a 'Closed Set' assumption, and there is seldom any work on OSR (Open Set Recognition) in federated settings. Such an assumption fails in realistic environments, where models frequently encounter classes unseen during training. In safety-critical applications like disease detection, the inability to identify these open-set samples poses severe risks of misdiagnosis in safety-critical healthcare applications.

Existing attempts to adapt OSR to federated settings often struggle with inter-client interference and inconsistent convergence. We take a divergent approach, hypothesized on the premise that global OSR performance is strictly bounded by the quality of local representations. Simply put, if the local client achieves a sufficiently robust closed-set decision boundary, the aggregated global model will inherently inherit this capability without requiring specialized open-set aggregation protocols.

Building on the insights of (Vaze et al., 2021) and contrastive anchor-based methodologies (Li et al., 2024), we propose **FedCPR** — a framework that enforces strict feature separation at the client level. We employ contrastive learning and, alongside, present two key technical innovations:

- A Latent Prototype Rejection (LPR) module, which generates perturbations in the feature space rather than the image space to simulate difficult out-of-distribution samples, and
- A Curriculum Learning strategy that guides the optimization trajectory from easy to hard examples. Together, these mechanisms force the client models to learn hyper-robust features that resist open-set misclassification, simplifying the subsequent global aggregation task.

2. Related Work

The evolution of Open Set Recognition (OSR) has shifted from basic thresholding of softmax scores to advanced representation learning. In the context of Federated Learning (FL), this challenge is intensified by data heterogeneity and the risk of parameter misalignment. Our work, FedCPR, builds upon several key advancements in both centralized OSR and federated learning. We explore state-of-the-art OSR methods at the client side and explore novel server aggregation methods to determine the best approach for open-set recognition in a federated setting.

PROSER performs OSR by introducing negative "placeholders" during training time (Zhou et al., 2021). These placeholders are generated at the latent level using manifold mixup, where features from different known classes are linearly combined to anticipate the distribution of potential unknown samples (Zhou et al., 2021). A dummy classification head is reserved to act as the boundary between known and unknown classes. This dummy head is tuned to produce high scores for mixed-up "unknown" latents and low scores for legitimate known samples by using an additional loss term (Zhou et al., 2021).

ARPL (Adversarial Reciprocal Points Learning) removes the traditional linear classification layer with reciprocal points for each known class (Chen et al., 2021). Each reciprocal point represents the "extra-class space" for a specific category; training aims to push class samples away from their respective reciprocal points (Chen et al., 2021). To prevent the latent space from expanding infinitely, ARPL introduces an adversarial margin constraint and generates confusing adversarial samples to further refine the boundary between known and unknown distributions (Chen et al., 2021).

Contrastive Learning for OSR Contrastive learning is powerful for learning discriminative feature spaces by pulling positive pairs together and pushing negatives apart. In the context of OSR, **Contrastive Class Anchor Learning (CCAL)** (Li et al., 2024) shifts the focus from instance-level comparisons to class-specific anchors. By maximizing the similarity between examples and their respective class-wise prototypes while ensuring adequate separation from the "open" regions of the hypersphere, CCAL creates compact known-class territories (Li et al., 2024).

Similarly, **Contrastive Open Set Recognition (ConOSR)** (Xu et al., 2023) utilizes a supervised framework (SupConST) that integrates soft targets generated via mixup and label smoothing. By contrasting real training examples against semantically ambiguous "virtual" samples, the model learns to focus on the most discriminative features of known classes. This effectively reduces the difficulty of outlier detection

by ensuring that such class-specific features are absent in unknown samples, supporting the "familiarity hypothesis" (Dietterich & Guyer, 2022).

FedPD (Federated Parameter Disentanglement) is a specialized framework designed to mitigate cross-client interference and parameter misalignment in FedOSR (Yang et al., 2023). It characterizes neurons based on task-related importance derived from the gradients of both closed-set and open-set losses (Yang et al., 2023). Specifically, it utilizes Local Parameter Disentanglement (LPD) to decouple client models into three non-overlapping components: close-specific, open-specific, and shared parameters (Yang et al., 2023). On the server side, Global Divide-and-Conquer Aggregation (GDCA) employs Optimal Transport (OT) to align corresponding parameter groups across heterogeneous clients before averaging, preventing the model collapse often seen in standard aggregation (Yang et al., 2023).

FedAvg (McMahan et al., 2023) is the standard baseline that performs a simple weighted average of client weights to aggregate the global model. While communication-efficient, it can struggle with the high parameter variance inherent in open-set recognition tasks on non-IID data.

Our proposed **FedCPR** builds on these principles by employing latent-level transformations to learn underlying structural features without the computational instability of generative adversaries.

3. The Proposed FedCPR

As outlined in Figure 1, our method focuses on specialized client training to address the FedOSR problem while utilizing the original FedAvg (McMahan et al., 2023) for server-side model aggregation. Specifically, we employ Supervised Contrastive Learning (SupCon) (Khosla et al., 2021) between the known classes for a client. For a batch of N samples, let $i \in \{1, \dots, N\}$ be the index of a sample and $P(i)$ be the set of indices of all other samples in the batch sharing the same class label. The loss is defined as:

$$\mathcal{L}_{SC} = \sum_{i=1}^N \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{j \neq i} \exp(z_i \cdot z_j / \tau)}$$

In this formulation, z represents the L_2 -normalized feature embedding and τ is the temperature hyperparameter. This encourages the local model to maximize the similarity between embeddings of samples from the same class while minimizing the similarity between the embeddings of all other known classes. Additionally, we introduce the novel Latent Prototype Rejection (LPR) which trains the model to minimize the Kullback-Leibler (KL) Divergence between curriculum-augmented intermediate features from known classes and a uniform distribution.

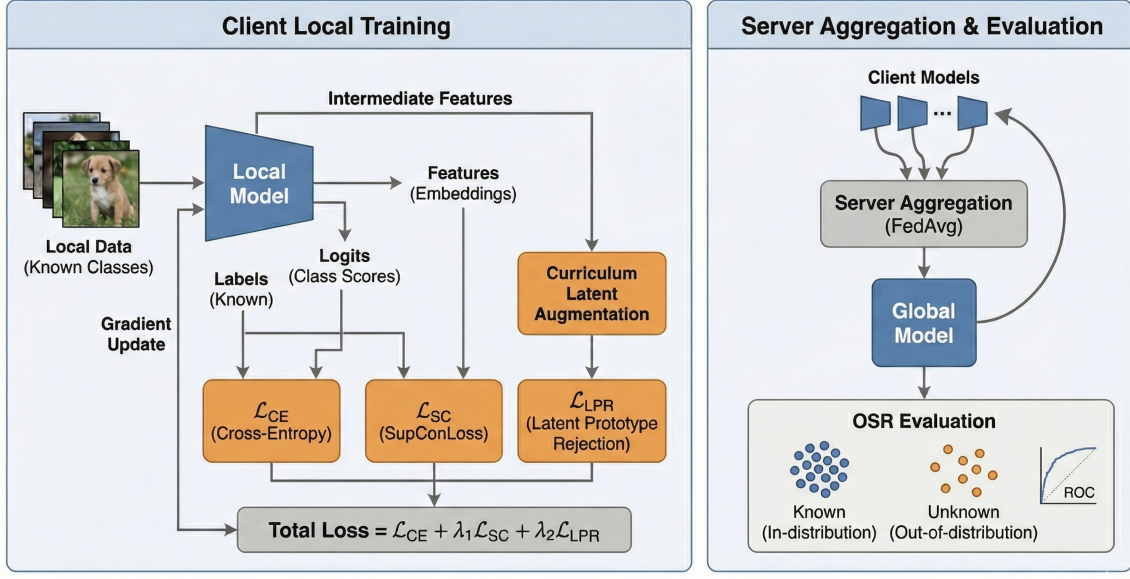


Figure 1. FedCPR Framework - Federated Contrastive and Prototype Rejection-based Open Set Recognition

Our method uses class centroids combined with energy-based scoring (Liu et al., 2021) for open-set detection, where samples with higher energy indicate higher uncertainty in the model and thus may be recognized as open-set. Thus, our model’s classifier head requires only K outputs, where K is the number of known classes.

3.1. Latent Prototype Rejection

We address the absence of negative (open-set) samples during local training through Latent Prototype Rejection (LPR). LPR performs curriculum-based feature-level augmentations to simulate “pseudo-open” samples. We split the local model $\mathcal{F}$ into two components: a feature extractor $\mathcal{F}_{pre}$ that maps input x to an intermediate latent space, and $\mathcal{F}_{latter}$ which maps this latent representation to the final normalized embedding z . For each batch of known samples, we extract intermediate features $h = \mathcal{F}_{pre}(x)$. We then apply a **Curriculum Latent Augmentation** strategy to h , where the complexity of the transformation increases with the global training round r . Specifically, T is the set of transformations $\{T_{jigsaw}, T_{flip}, T_{rotate}\}$. The chosen transformation $T(h, r)$ disrupts the spatial structure of the latent features to create a “pseudo-open” representation h' .

To ensure the model learns to reject these synthetic outliers, we pass h' through the remainder of the network to obtain $z' = \mathcal{F}_{latter}(h')$. We then compute the logits for these trash samples against the class prototypes (anchors) W stored in the final linear layer. The LPR loss is defined as the KL divergence between the predictive distribution of these trash

samples and a uniform distribution $\mathcal{U}$:

$$L_{LPR} = D_{KL} \left(\text{Softmax} \left(\frac{z' \cdot W^T}{\tau} \right) \parallel \mathcal{U} \right)$$

where τ is a temperature hyperparameter. By forcing the model to produce a uniform distribution for structurally disrupted features, we effectively hollow out the decision space between known class clusters, leading to more effective rejection of true open-set samples.

3.2. Combined Objective

The local training at each client k is governed by our loss function designed to balance class discriminability, feature compactness, and outlier rejection. Given a local dataset $\mathcal{D}_k$, the client minimizes:

$$\min_{\theta} \mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda_{sc} \mathcal{L}_{SupCon} + \lambda_{lpr} \mathcal{L}_{LPR}$$

where λ_{sc} and λ_{lpr} are hyperparameters controlling the influence of the contrastive and rejection components, respectively. $\mathcal{L}_{CE}$ ensures high classification performance on known classes, $\mathcal{L}_{SupCon}$ enforces tight intra-class clustering, and $\mathcal{L}_{LPR}$ reserves low-probability regions for unknown samples.

The client training procedure for FedCPR is shown in Algorithm 1.

3.3. Energy-based Detection

For the inference phase, we leverage the learned class prototypes to distinguish between known and unknown samples.

Table 1. Performance of our method and baselines on CIFAR-10 with 6 known classes and 4 open-set classes.

Method	Closed-set ACC (%)	Open-set AUROC (%)
PROSER (Zhou et al., 2021)	90.62 $\pm$ 3.35	80.77 $\pm$ 2.86
ARPL (Chen et al., 2021)	90.00 $\pm$ 3.98	73.41 $\pm$ 4.28
FedPD (Yang et al., 2023)	90.79 $\pm$ 3.16	80.58 $\pm$ 2.50
SupCon (Khosla et al., 2021)	89.93 $\pm$ 3.74	80.83 $\pm$ 1.82
FedCPR (ours)	90.96 $\pm$ 2.82	85.42 $\pm$ 4.02

In FedCPR, the weights $W \in \mathbb{R}^{K \times D}$ of the final layer serve as K class centroids. Given a test sample x , we first extract its normalized embedding $z = \mathcal{F}(x)$. While distance-based metrics like cosine similarity to the nearest centroid provide a baseline for detection, they often fail to capture the global uncertainty of the model. Therefore, we adopt an Energy-based scoring mechanism. We define the logit for class k as $g_k(z) = (z \cdot w_k)/\tau$. The energy score $E(z, \tau)$ is then computed as:

$$E(z, \tau) = -\tau \cdot \log \sum_{k=1}^K \exp(g_k(z))$$

In this framework, a sample x is classified as "Open-Set" if $E(z, \tau) > \delta$, where δ is a set threshold. Higher energy values correspond to lower probability density in the feature space, signifying that the sample does not align strongly with any of the known class prototypes. This energy-based approach provides a robust indicator of "unknownness" than raw softmax probabilities, which are susceptible to high-confidence and overgeneralization.

Algorithm 1 FedCPR Client Operations

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1: Inputs: Aggregated model  $\theta_n$ , Local epochs  $E$ , learning rate  $\eta$ , current global round  $\rho$ , hyperparameters  $\lambda_{sc}, \lambda_{lpr}, \tau$ 
2: function ClientUpdate( $\theta_n, E, \eta, \rho, \lambda_{sc}, \lambda_{lpr}, \tau$ )
3:    $\theta_n \leftarrow \theta$ 
4:   for each local epoch  $e = 1, \dots, E$  do
5:     for each batch  $(x, y) \in \mathcal{D}_n$  do
6:        $logits, z \leftarrow \mathcal{F}(x; \theta_n)$  {Forward Pass}
7:        $\mathcal{L}_{known} \leftarrow \mathcal{L}_{CE}(logits, y) + \lambda_{sc} \mathcal{L}_{SupCon}(z, y)$ 
8:       {Latent Prototype Rejection}
9:        $h \leftarrow \mathcal{F}_{pre}(x; \theta_n)$ 
10:       $h' \leftarrow \text{CurriculumAug}(h, \rho)$  {Apply Tier 1, 2, or 3 transformations}
11:       $z' \leftarrow \mathcal{F}_{latter}(h'; \theta_n)$ 
12:       $logits_{aug} \leftarrow \frac{z' \cdot W^T}{\tau}$ 
13:       $\mathcal{L}_{LPR} \leftarrow D_{KL}(\text{Softmax}(logits_{aug}) \parallel \mathcal{U})$ 
14:       $\theta_n \leftarrow \theta_n - \eta \nabla(\mathcal{L}_{known} + \lambda_{lpr} \mathcal{L}_{LPR})$ 
15:    end for
16:  end for  $\theta_n$ 
17: end function

```

4. Experiment and Results

We evaluate our method alongside other leading OSR and FedOSR methods in a federated setting using the CIFAR-10 dataset with $K = 6$ known classes and the remainder 4 classes as the open-set. We train 5 clients for 50 global rounds. Each client performs 5 local epochs per global round, totaling 250 local epochs per client. We utilize the SGD optimizer with momentum= 0.9 and weight decay= $5e - 4$. We use a batch size of 128, learning rate of 0.02, λ_{SC} of 0.5 and λ_{lpr} of 1.0. The temperature for SupCon loss and energy scoring is set to $\tau = 0.08$. We use a 8-layer CNN model. Experiments are performed on a Google T4 GPU using PyTorch.

4.1. Performance Analysis

The experimental results in Table 1 demonstrate that our proposed **FedCPR** framework consistently outperforms existing state-of-the-art methods in identifying unknown classes while maintaining competitive closed-set classification accuracy. FedCPR achieves the highest overall performance with a best open AUROC of 85.42% and a best closed-set Accuracy of 90.96%.

Superior Outlier Detection Compared to specialized OSR frameworks like PROSER (Zhou et al., 2021) (80.77%) and FedPD (Yang et al., 2023) (80.58%), FedCPR yields an improvement of approximately 5%. This suggests that our latent-level curriculum-based-augmentation strategy is more effective at defining the "known" boundary than previous generative or parameter-disentanglement approaches.

Instability of Generative Baselines Interestingly, ARPL (Chen et al., 2021) exhibited the lowest AUROC (73.41%) in our testing. This performance degradation likely stems from the inherent instability of training reciprocal points and generative adversaries in a decentralized environment where global aggregation can disrupt finely-tuned adversarial boundaries.

4.2. The Role of the Familiarity Hypothesis

The robust performance of FedCPR can be theoretically explained by the **Familiarity Hypothesis** (Dietterich & Guyer, 2022; Xu et al., 2023). This hypothesis suggests that deep

OSR models detect unknown samples by identifying the **absence** of necessary features required to categorize a sample as "familiar" rather than recognizing novel features (Xu et al., 2023; Dietterich & Guyer, 2022).

By utilizing a supervised contrastive objective, FedCPR ensures that the backbone learns highly discriminative features focused on the most distinctive patterns of known classes. Because these learned features are so specific to the known training domain, they are statistically unlikely to appear in unknown test samples. This increased feature specificity makes the "absence" of such patterns easier to detect, thereby improving the model's rejection capabilities.

Table 2. Effect of removing or modifying key components of our method on open-set performance. *A non-adaptive curriculum for latent augmentation is used

Method	Open-set AUROC (%)
Ours without LPR	82.88
Ours without curriculum*	88.23
Ours without SupCon	88.46
Ours with image augmentation	89.00
FedCPR (ours)	89.89

4.3. Ablation of FedCPR components

The efficacy of FedCPR is further validated through an ablation study focusing on its core components (Table 2):

- **Latent Prototype Rejection (LPR):** Removing the LPR mechanism (equivalent to SupCon) results in a significant AUROC drop from 89.89% to 82.88%, underscoring the necessity of specialized OSR-targeted inputs/latents for the model to learn from. Additionally, this highlights that the SupCon approach alone is insufficient in the FedOSR setting.
- **Adaptive Curriculum:** Omitting the curriculum-based complexity shift reduces performance to 88.23%. This confirms that progressively increasing the severity of latent corruption (e.g., Jigsaw and rotations) forces the backbone to internalize deep structural representations rather than superficial shortcuts.
- **SupCon:** Removing the SupCon optimization term from the objective leads to a 1.43% in AUROC, suggesting that explicitly encouraging tight clusters with well-defined boundaries enhances open-set performance. This follows from literature which suggests that closed-set approaches may be "all you need" for enhanced open-set performance (Vaze et al., 2021).
- **Image Augmentation:** Applying LPR on closed-set images to generate "pseudo-open" samples instead of doing so with intermediate latents reduces AUROC by

0.89%, pointing towards the superior representations of "openness" achieved when using latent-based strategies. This highlights why previous methods such as PROSER (Zhou et al., 2021) achieved superior performance over GAN-based image generation by focusing on latent-mixups.

4.4. Latent Space Visualization

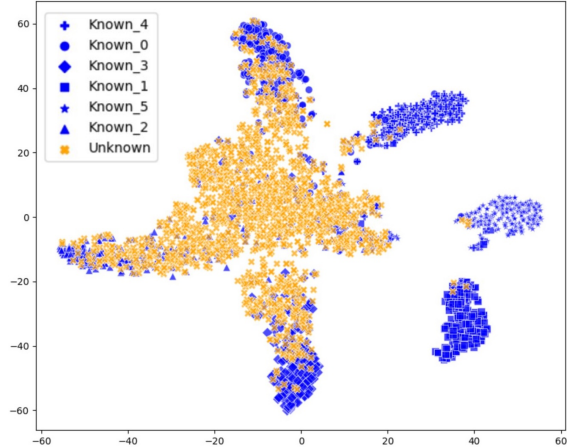


Figure 2. t-SNE Plot showing separation of closed and open set class embeddings when using FedCPR

The t-SNE visualization The t-SNE plot (Van der Maaten & Hinton, 2008) (Figure 2) provides further empirical evidence of this structural learning. The known classes are represented by highly compact, well-separated clusters, while the "Unknown" samples are widely dispersed throughout the open space. Because FedCPR pulls known samples toward class anchors (Li et al., 2024) while pushing "prototype" latents into the open space, it creates clear "empty" regions in the latent manifold, facilitating reliable outlier rejection during evaluation, especially when using the energy-based scoring method.

5. Discussion

Beyond its empirical performance, **FedCPR** offers significant practical advantages in terms of computational savings and system-level robustness. By working with intermediate latent representations rather than employing pixel-level generative models, our method greatly reduces the local compute overhead, making it very well-suited for resource-constrained edge devices. Furthermore, the framework maintains superior communication efficiency by utilizing the standard FedAvg protocol. Unlike specialized methods like **FedPD** that require the transmission of parameter-disentangled sub-components or auxiliary gradients, **FedCPR** achieves high performance with minimal network strain and no additional bandwidth require-

ments. The implementation stability is another key highlight, as our curriculum-based strategy avoids the volatile optimization trajectories and gradient conflicts often associated with adversarial training. These findings suggest that well-structured latent-space perturbations can achieve superior open-set rejection with reduced hyperparameters and without training instability of previous FedOSR works.

6. Conclusion

In this work, we address the challenge of Federated Open-Set Recognition (FedOSR), where models must identify unknown classes without access to real outliers. To enable robust detection in this constrained setting, we propose FedCPR, a novel framework leveraging Latent Prototype Rejection. Specifically, on the client side, we introduce a curriculum-based augmentation strategy that perturbs intermediate feature maps (rather than raw pixels) to generate synthetic outliers. These latent prototypes are explicitly pushed away from known class prototypes to create precise decision boundaries without conflicting with closed-set learning. Extensive experiments on federated benchmarks demonstrate that FedCPR effectively resolves the semantic validity paradox of pixel augmentation and achieves state-of-the-art performance in open-set detection. Future research can extend this latent-space approach by incorporating Adversarial Training, specifically applying FGSM to augmented latents to generate "worst-case" outliers that lie on the manifold's edge. Additionally, exploring adaptive curriculum strategies and alternative feature perturbations, offers a promising direction for further optimizing robustness against diverse open-set scenarios.

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